

Cost-Push Markup Shocks in the Rubbo Network Phillips Curve

1 Baseline Objects

Let there be N sectors. The calibrated primitives are

$$\alpha \in \mathbb{R}^N, \quad \beta \in \mathbb{R}^N, \quad \Omega \in \mathbb{R}^{N \times N}, \quad \Delta = \text{diag}(\delta_i).$$

Here α_i is the labor share in sector i , β_i is the final-consumption weight, Ω_{ij} is the input share of good j in sector i , and δ_i is the Calvo price-adjustment probability.

The Domar/sales exposure vector is

$$\lambda' = \beta'(I - \Omega)^{-1}.$$

Define the rigidity-adjusted Leontief inverse

$$R_\Delta = (I - \Omega\Delta)^{-1},$$

and the common denominator

$$d = 1 - \beta'\Delta R_\Delta \alpha.$$

The sectoral Phillips-curve slope vector, following Rubbo's Matlab convention, is

$$b = (\gamma + \varphi) \frac{\Delta R_\Delta \alpha}{d}.$$

The CPI Phillips-curve slope is

$$\kappa^C = \beta'b.$$

2 Productivity-Shock Term in the Baseline Model

For reference, the productivity-shock propagation matrix used in the original code is

$$v^A = \Delta R_\Delta \left[\alpha \frac{\lambda' - \beta'\Delta R_\Delta}{d} - I \right].$$

The CPI residual generated by productivity shocks is

$$u_t^C = \beta'v^A d \log A_t.$$

3 Markup Cost-Push Shocks

Following the supplement, introduce an exogenous desired-markup shock

$$\mu_t^D \in \mathbb{R}^N.$$

A positive element of μ_t^D is a cost-push shock in that sector. A sectoral subsidy can be represented as a negative markup shock:

$$\mu_t^{D,\text{net}} = \mu_t^{D,\text{shock}} - s_t.$$

The consumer-price markup term is

$$v_t^C = \frac{\beta' \Delta (I - \Omega \Delta)^{-1}}{1 - \beta' \Delta (I - \Omega \Delta)^{-1} \alpha} \mu_t^D.$$

This is the object computed in the static module as `CPI_markup`.

The divine-coincidence markup term is

$$\pi_t^{DC} = \rho E_t \pi_{t+1}^{DC} + (\gamma + \varphi) \tilde{y}_t + \lambda' \mu_t^D.$$

Therefore, absent expectations and with $\tilde{y}_t = 0$, the static DC response is

$$\lambda' \mu_t^D.$$

The output gap that stabilizes the divine-coincidence index on impact is

$$\tilde{y}_t^{DC=0} = -\frac{\lambda' \mu_t^D}{\gamma + \varphi}.$$

4 Sectoral Phillips Curves

Let

$$B = \frac{\Delta R_\Delta \alpha}{d}$$

so that $b = (\gamma + \varphi)B$. Rubbo's sectoral matrix is

$$V = [\Delta R_\Delta - B (\lambda' - \beta' \Delta R_\Delta)] (I - \Omega).$$

The sectoral Phillips curves with desired-markup shocks are

$$\pi_t = \rho(I - V)E_t \pi_{t+1} + b\tilde{y}_t - V\chi_t + (I - V)\Delta(I - \Delta)^{-1}\mu_t^D.$$

In the current markup-shock module, the productivity term is set to zero, $\chi_t = 0$, so the dynamic recursion is

$$\pi_t = \rho(I - V)E_t \pi_{t+1} + b\tilde{y}_t + (I - V)\Delta(I - \Delta)^{-1}\mu_t^D.$$

5 Dynamic Experiment

The current dynamic script assumes an exogenous AR(1) process:

$$\mu_{t+1}^D = \rho_\mu \mu_t^D.$$

Given a terminal condition $\pi_{T+1} = 0$, the sectoral Phillips curve is solved backwards:

$$\pi_t = \rho(I - V)\pi_{t+1} + b\tilde{y}_t + (I - V)\Delta(I - \Delta)^{-1}\mu_t^D.$$

The reported CPI inflation response is

$$\pi_t^C = \beta' \pi_t,$$

and the DC response is computed from the scalar recursion

$$\pi_t^{DC} = \rho\pi_{t+1}^{DC} + (\gamma + \varphi)\tilde{y}_t + \lambda'\mu_t^D.$$

6 Interest-Rate Implementation

The output-gap rules above can be translated into nominal interest-rate paths using Rubbo's log-linear IS curve. With consumer-price inflation entering the household Euler equation, the IS curve is

$$\tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\gamma}(i_{t+1} - E_t\pi_{t+1}^C - r_{t+1}^{nat}).$$

Rearranging gives the policy-rate response relative to the natural nominal rate:

$$i_{t+1} - r_{t+1}^{nat} = E_t\pi_{t+1}^C + \gamma(E_t\tilde{y}_{t+1} - \tilde{y}_t).$$

The code uses the simulated future paths as the model-consistent expectations:

$$E_t\pi_{t+1}^C = \pi_{t+1}^C, \quad E_t\tilde{y}_{t+1} = \tilde{y}_{t+1},$$

with terminal values set to zero at the end of the finite horizon. Therefore the computed interest-rate path is

$$\hat{i}_{t+1} \equiv i_{t+1} - r_{t+1}^{nat} = \pi_{t+1}^C + \gamma(\tilde{y}_{t+1} - \tilde{y}_t).$$

Because the oil experiment is a desired-markup shock rather than a productivity shock, the productivity component of r_{t+1}^{nat} is unchanged. Thus $\hat{i}_{t+1}$ is the nominal policy-rate movement above the natural-rate path. When reporting peak interest-rate movements, the code excludes the final finite-horizon transition by default, because the last point may reflect the imposed terminal condition rather than the economic response. The full path is still stored.

7 Policy Rules Implemented

The first dynamic module implements two simple output-gap rules:

$$\tilde{y}_t = 0 \quad \text{or} \quad \tilde{y}_t = -\frac{\lambda'\mu_t^D}{\gamma + \varphi}.$$

The first isolates pass-through and inflation dynamics. The second stabilizes the divine-coincidence Phillips curve period by period. A full Taylor-rule block can be added later by combining the markup-shock recursion with Rubbo's monetary IRF system.

8 Policy Comparison Module

The policy-comparison script expresses monetary policy as a path for the output gap. This is a reduced-form policy experiment: it asks what output-gap movement would be needed under alternative targets, before translating that movement into an interest-rate rule through an IS curve.

The rules currently implemented are:

$$\begin{aligned}\tilde{y}_t^0 &= 0, \\ \tilde{y}_t^{DC} &= -\frac{\lambda' \mu_t^D}{\gamma + \varphi},\end{aligned}$$

and a CPI-stabilizing rule. Let π_{t+1} denote the sectoral inflation vector already computed in the backward recursion. The CPI-stabilizing output gap is chosen so that $\beta' \pi_t = 0$:

$$\tilde{y}_t^{CPI} = -\frac{\rho \beta' (I - V) \pi_{t+1} + \beta' (I - V) \Delta (I - \Delta)^{-1} \mu_t^D}{\beta' b}.$$

The script also includes partial-reaction rules:

$$\tilde{y}_t^{lean, CPI} = \theta_C \tilde{y}_t^{CPI}, \quad \tilde{y}_t^{lean, DC} = \theta_{DC} \tilde{y}_t^{DC},$$

where $\theta_C, \theta_{DC} \in [0, 1]$.

The reported losses are transparent quadratic losses, not the full welfare matrix from Rubbo's Section 5:

$$\begin{aligned}\mathcal{L}^{CPI} &= \sum_{t=0}^{T-1} \rho_L^t [(\pi_t^C)^2 + \omega_y \tilde{y}_t^2], \\ \mathcal{L}^{DC} &= \sum_{t=0}^{T-1} \rho_L^t [(\pi_t^{DC})^2 + \omega_y \tilde{y}_t^2],\end{aligned}$$

and

$$\mathcal{L}^{dual} = \sum_{t=0}^{T-1} \rho_L^t [\omega_C (\pi_t^C)^2 + \omega_{DC} (\pi_t^{DC})^2 + \omega_y \tilde{y}_t^2].$$

These losses are intended to compare simple central-bank responses to a markup shock: ignore the shock, stabilize CPI, stabilize the divine-coincidence index, or lean partially against either target.

9 Code Procedures

The project is organized in layers. The prototype files are left unchanged. The new markup-shock layer adds:

1. `markup_costpush_static.m`: computes static pass-through objects, including CPI markup response, divine-coincidence response, the output gap that stabilizes divine-coincidence inflation, network pressure, and the sectoral Phillips-curve markup term.
2. `markup_costpush_irf.m`: builds an AR(1) path for μ_t^D and solves the forward-looking sectoral Phillips curves backward over a finite horizon.
3. `markup_costpush_policy_compare.m`: compares simple output-gap policy rules and computes transparent quadratic losses.

4. `markup_policy_rate_path.m`: converts each output-gap path into a nominal interest-rate path relative to the natural rate using the IS curve above.

The main drivers are:

1. `run_oil_markup_module_example.m`: static and dynamic pass-through under a baseline oil markup shock and an illustrative subsidy.
2. `oil_markup_monetary_policy.m`: compares output-gap policy rules, inflation responses, output-gap paths, quadratic losses, and interest-rate paths.
3. `oil_markup_interest_rate_policy.m`: focuses on the interest-rate implementation of the same policy rules.

All drivers follow Rubbo-style paths and can be run from the replication root, from `calibration/oil shocks`, or from `calibration/data cleaning/IO tables`.